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Cybersecurity and Cyber-Resilience for the European Cloud–Edge–IoT Computing Continuum: A Strategic Capability Roadmap

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ABSTRACT The distribution of computation across cloud, edge, and IoT tiers, combined with the simultaneous operationalisation of NIS2, the Cyber Resilience Act, the AI Act, and DORA, has made cybersecurity, resilience, and digital sovereignty an integrated strategic problem for Europe. Perimeter-oriented security models do not scale to heterogeneous, physically exposed, and multi-jurisdictional continuum infrastructures. This paper addresses that problem by proposing a three-stage methodology for identifying and prioritising cybersecurity capabilities: architectural decomposition into six Security Layers and four Transversal Cross-Layers, seven-dimension capability profiling, and priority clustering with phased transition matrices. Applying the methodology, the paper identifies ten strategic capabilities spanning hardware roots of trust, post-quantum cryptographic agility, lightweight edge security, federated interoperability, supply chain verification, cognitive security operations, pan-European network federation, confidential computing, resilient infrastructure, and proactive threat management. These are consolidated into three executive priority clusters, Sovereign Trusted Infrastructure, Cognitive Security and Autonomous Cyber Defence, and Federated Operational Sovereignty, analysed over a 2026–2040 transition horizon. The contribution is both methodological, providing a reproducible template for cybersecurity roadmap construction, and strategic, framing autonomous, federated, and certified Security-by-Design as the operational target for European digital infrastructure.

INDEX TERMS Certification, Cloud–Edge–IoT continuum, confidential computing, cyber-resilience, digital sovereignty, European Union, post-quantum cryptography, security roadmap, Zero-Trust architecture.

I. INTRODUCTION

The shift from centralised cloud to a distributed Cloud–Edge–IoT continuum relocates computation, storage, and inference closer to devices, users, and physical processes [1]–[3]. The architectural benefit is lower latency and higher operational autonomy; the security consequence is a sharp increase in attack surface, heterogeneity, and the number of administrative and trust domains that a defender must reason about simultaneously. Perimeter-oriented and siloed security models, designed for consolidated data-centre environments, are not structurally adequate for infrastructures that are geographically dispersed, resource-constrained, and physically exposed [4], [5].

Europe faces this transition under four simultaneous regulatory activations. NIS2 [6], the Cyber Resilience Act (CRA) [7], the AI Act [8], and DORA [9] together raise the baseline of mandatory security, resilience, accountability, and supply chain obligations across essential entities, ICT products, high-risk AI systems, and financial sector third-party providers. The EU cybersecurity certification framework [10], [11] provides the assurance substrate these regulations presuppose. Operationalisation of this combined regime is uneven: national transposition lags the legal timetable in several large Member States [12], EUCS remains unadopted [13], and conformity assessment infrastructure for AI-specific security obligations is not yet in place [14].

The central argument of this paper is that Europe's normative leadership in trust, privacy, governance, and certification [15], [16] translates into strategic advantage only when it is embedded in deployable capabilities. We call the distance between normative leadership and deployable capacity the ambition–implementation gap. Closing this gap requires a transition from reactive, siloed security towards autonomous, federated, and certified Security-by-Design: infrastructures that sustain trustworthy operation under adverse conditions (autonomous), security mechanisms that span providers and jurisdictions (federated), and assurance embedded through the system lifecycle (certified Security-by-Design) [4], [17], [18].

The paper makes four contributions. First, it frames continuum cybersecurity as an interlocking problem across six Security Layers and four Transversal Cross-Layers (Section III). Second, it defines a three-stage methodology linking architectural decomposition to European strategic concerns through a Sovereignty Level construct (Section IV). Third, it identifies ten strategic capabilities, consolidated into three priority clusters analysed over a 2026–2040 transition horizon (Sections V–VI). Fourth, it derives implementation-oriented recommendations connecting research, policy, industrial capacity, and sovereignty (Section VII).

Section II provides the strategic and regulatory context. Section III defines the system model. Section IV presents the methodology. Section V identifies focus areas. Section VI presents the three priority clusters and their transition matrices. Section VII states the recommendations. Section VIII discusses validation and limitations. Section IX concludes.

II. BACKGROUND AND DRIVING FACTORS

The continuum is characterised by decentralisation and physical exposure. Decentralisation complicates visibility, orchestration, patching, identity, and incident response across multiple administrative domains and increases the probability of inconsistent security postures across deployments [19], [20]. Physical exposure removes the environmental and access controls that hyperscale data centres rely on: edge nodes in factories, substations, transport hubs, and telecom sites are exposed to tampering, fault injection, side-channel attacks, and supply chain compromise [21]. Hardware roots of trust, secure boot, remote attestation, and tamper-aware design are baseline requirements rather than optional enhancements [22], [23].

The European regulatory activation is historically dense. NIS2 [6], the CRA [7], the AI Act [8], DORA [9], the Data Act [24], and GDPR [25] together impose obligations that must be translated into architectures, assurance mechanisms, and lifecycle controls operating across cloud, edge, and IoT. The simultaneity burdens SMEs and smaller Member States disproportionately.

Sovereignty is the third vector. Dependence on non-EU providers for secure silicon, cloud platforms, and enterprise security tooling creates geopolitical and operational risk. Extra-territorial data access under foreign jurisdiction has

Security framework schematic omitted in this build variant.

It contains four execution environments (HPC, Cloud, Edge, Devices), six Security Layers (L1–L6), and four Transversal Cross-Layers (CL1–CL4).

FIGURE 1. Security framework: four execution environments, six Security Layers (vertical stack, L1–L6), four Transversal Cross-Layers (CL1–CL4).

been confirmed as a practical constraint for EU users of hyperscaler services [26].

Europe's position in this landscape is asymmetric. A world-leading regulatory foundation and a rigorous certification framework [10], [11] coexist with structural industrial gaps. EU-headquartered vendors (Infineon, NXP, STMicroelectronics) design the dominant secure element products and achieved EUCC “High” certification for multiple product families in 2025 [27], but fabrication depends on non-EU foundries and designs remain proprietary, preventing sovereign auditability at the silicon level. Intel, AMD, and ARM control the operational Trusted Execution Environment (TEE) market. No EU-origin RISC-V processor has received EUCC or Common Criteria certification as of April 2026. The enterprise SIEM market is dominated by US platforms, with no EU-headquartered vendor in the Gartner Leaders quadrant as of late 2025 [28]. Hyperscalers hold approximately 70% of the EU cloud market; EU-headquartered providers collectively hold about 15%, stable since 2022 but below the 29% share recorded in 2017 [29]. For post-quantum migration the NIS Cooperation Group roadmap [30] now provides a binding framework, but production-scale post-quantum cryptographic (PQC) hardware tailored for edge and IoT is not yet available in Europe.

III. SYSTEM MODEL

The continuum is modelled as a layered reference architecture spanning four execution environments (HPC, Cloud, Edge, Devices) and organised into six Security Layers (vertical stack) and four Transversal Cross-Layers (horizontal enablers). Figure 1 shows the schema. The decomposition aligns with ETSI MEC GS 003 [31], NIST SP 800-207 and SP 800-207A [4], [5], NIST CSF 2.0 [32], ISO/IEC 27400 [33], ENISA's baseline for IoT [34], and the distributed continuum literature [1]–[3]. Layers are analytically distinct but operationally interdependent: a gap at one layer propagates to adjacent layers [20], [21].

A. SECURITY LAYERS (L1–L6)

L1 Hardware-Centric Security. Physical and silicon-level trust anchors: roots of trust, secure boot, remote attestation, TPMs, PUFs, and physical-layer detection primitives [22], [23]. Without a verifiable silicon anchor, higher-layer attestation collapses into unfounded assumptions. L1 is the principal locus of hardware sovereignty and the layer where

EUCC “High” certification [11] operates as the regulatory instrument of trust.

L2 Software-Centric Security. Hypervisors, container runtimes, distributed operating systems, and the update pipelines that maintain them. Central concerns are attack-surface reduction through hardening and cryptographically verified, failure-resilient software distribution to field-deployed nodes where physical remediation is infeasible [35], [36].

L3 Application-Centric Security. Containerised applications, serverless functions, service meshes, and the APIs that expose them. NIST SP 800-204, 204A, 204C [37]–[39] and the OWASP API Security Top 10 [40] provide the canonical reference controls; AI Act and GDPR Article 25 obligations surface here as runtime behaviours.

L4 User and Orchestration-Centric Security. Continuous, context-adaptive authorisation aligned with NIST SP 800-207 and 207A [4], [5]; federated and self-sovereign identity based on W3C Verifiable Credentials [41] and, in Europe, eIDAS 2 [42] with the EU Digital Identity Wallet implementing acts [43]; privileged-access controls that eliminate standing credentials.

L5 Data-Centric Security. Protection of data at rest, in transit, and in use. Confidential computing through TEEs [44]; PQC based on FIPS 203/204/205 [45]–[47]; signalling-data protection; distributed trust; Quantum Key Distribution for high-value links [48], [49]; Secure Multi-Party Computation for collaborative processing across distrusting parties.

L6 Network Infrastructure and Perimeter-Centric Security. 5G/6G cores, disaggregated RAN, IT/OT gateways, far-edge compute, and non-terrestrial segments [50]–[53]. Security integrates with physical resilience: digital breaches must not cascade into kinetic damage.

B. TRANSVERSAL CROSS-LAYERS (CL1–CL4)

CL1 Automated Security Processes and Cyber-Resilient AI Lifecycle. Secure-by-design development, runtime supply chain verification, and continuous adversarial validation, anchored in NIST SP 800-204C DevSecOps patterns [39], the NIST AI Risk Management Framework and its Generative AI Profile [54], [55], the SLSA provenance framework [56], and the adversarial robustness literature [57].

CL2 Dynamic Risk Management and Operational Cyber Resilience. Federated SOC operation [58], NWDAF-integrated analytics [59], LLM-based SecOps, and CTI lifecycle governance using MITRE ATT&CK [60], STIX and TAXII 2.1 [61], [62], and ENISA threat landscape reporting [63].

CL3 Interoperable Security Mechanisms. Federation anchored in Simpl [64], CAMARA Network APIs [65], semantic interoperability through OASC MIMs [66] and ETSI NGSI-LD [67], and federated cyber ranges.

CL4 Compliance-by-Design, Agile Governance, and Automated Assurance. Continuous certification (EUCC; EUCS when adopted) [10], [11] using NIST OSCAL [68], [69]; multicloud key management; privacy-preserving feder-

ated computation under GDPR Article 25; cyber risk quantification; and AI trustworthiness under AI Act Article 9 [8], [54].

IV. METHODOLOGY

The methodology transforms a heterogeneous strategic landscape into a structured roadmap through three stages followed by iterative validation. Two commitments shape it. The methodology supports strategic roadmap construction, not deterministic forecasting: in complex socio-technical ecosystems, progress is non-linear and depends on interacting technical, regulatory, and market developments. The methodology connects technical capabilities with a Sovereignty Level (SSL) construct that captures the degree to which critical capabilities, control points, and dependencies remain under credible European influence. SSL is an interpretive lens, not a scoring model.

A. STAGE 1: ARCHITECTURAL DECOMPOSITION

Stage 1 produces the structured map of the problem space used throughout the paper (Section III). Focus areas are derived from four sources: architectural characteristics of distributed cloud–edge–IoT systems, recurring cybersecurity challenges across HPC/Cloud/Edge/Device tiers, cross-cutting strategic concerns (sovereignty, trust, interoperability, resilience, assurance, regulatory alignment), and the operational need for credible deployment pathways. The output is a taxonomy of six Security Layers and four Cross-Layers within which sovereignty-sensitive focus areas are flagged for tracking across subsequent stages.

B. STAGE 2: CAPABILITY IDENTIFICATION AND PROFILING

A capability is a strategic ability the European ecosystem must be able to design, deploy, govern, standardise, assure, coordinate, or scale. Each capability is assessed through a seven-dimension profile applied identically across all priorities: (D1) Baseline and Target Objective with quantified end-states; (D2) Primary Drivers, separating threat drivers from regulatory drivers with cited articles and enforcement dates; (D3) High-Impact Activities, three to five interventions each tagged with their primary phase; (D4) Catalysts and Enablers, naming funding instruments, standards bodies, and ecosystem levers; (D5) Disruptors and Systemic Risks, covering technical cliffs, supply chain risks, and geopolitical risks; (D6) Failsafe Mechanisms, specifying engineering fallbacks and graceful degradation; (D7) Transversal Meta-Layer, connecting to sovereignty, trust and ethics, and sustainable security. The schema enables structured cross-priority comparison and exposes shared dependencies and bottlenecks.

C. STAGE 3: PRIORITY CLUSTERING AND TRANSITION MATRICES

Capabilities that share architectural proximity, enabling conditions, or strategic outcomes are grouped into Priority Clusters. Each cluster is analysed through a three-phase transition

matrix: Phase 1 (2026–2030) Near-Term Implementation and Pilot Actions; Phase 2 (2030–2035) Mid-Term Scale-Up, Industrial R&D, and Standardisation; Phase 3 (2035–2040) Long-Term Strategic Autonomy and Leadership. For each cluster and phase, four analytical perspectives are combined: R&I maturity, policy/regulation/standards, operational deployment and adoption, and market uptake and strategic impact. Progression is expressed through transition conditions, evidence-informed readiness markers rather than deterministic gates, reflecting the non-linear nature of the underlying developments.

Two fixed vocabularies structure the transition matrices. The Operational Deployment Phase (ODP) scale runs Applied R&D → Fragmented Piloting → Standardised Integration → Ubiquitous Operation. The Strategic Sovereignty Level (SSL) scale runs High Foreign Dependency → Emerging EU Ecosystem → Full Sovereign Autonomy.

V. FOCUS AREAS

Table 1 consolidates the focus areas per layer. Each is the atomic unit from which the capability profiles of Section VI are derived. The table replaces an exhaustive per-layer narrative; layer-level context is provided in Section III.

VI. PRIORITY CLUSTERS AND TRANSITION MATRICES

The ten capabilities are consolidated into three priority clusters, each presented through the seven-dimension schema with per-priority sub-entries, followed by a three-phase transition matrix.

A. CLUSTER 1: SOVEREIGN TRUSTED INFRASTRUCTURE

Cluster 1 unifies P1 (Hardware Roots of Trust), P2 (PQC Agility), P3 (Lightweight Cryptography and Physical-Layer Security), P8 (Confidential Computing), and P9 (Resilient-by-Design Infrastructure and Island Mode). Federated policies and autonomous defence gain operational credibility in proportion to the maturity of the execution substrate. The substrate must guarantee hardware integrity through open-source silicon roots of trust, quantum-resistant confidentiality through crypto-agile schemes aligned with FIPS 203/204/205, and autonomous resilience under network or orchestration-plane compromise. The shared critical dependency is European sovereign fabrication capacity under the EU Chips Act [75], [76].

a: D1 Baseline and Target.

P1: Infineon, NXP, and STMicroelectronics achieved EUCC “High” for proprietary secure element families in 2025 [27]; fabrication remains non-EU. No EU-origin RISC-V processor has received EUCC or Common Criteria certification. Target: $\geq 60\%$ of critical infrastructure edge nodes use EU-designed, EUCC “High” certified secure elements by 2040. P2: Edge and IoT environments depend on classical algorithms. The NIS CG roadmap [30] mandates national roadmaps by December 2026, high-risk critical infrastructure

transitioned by December 2030, full completion by December 2035. Target: quantum-native continuum with full cryptographic agility. P3: Extreme-edge devices (mMTC, body-area, backscatter) are constrained in compute, energy, and memory; conventional cryptography is impractical. Target: lightweight, adaptive security mechanisms, including zero-energy physical-layer authentication. P8: Intel TDX, AMD SEV-SNP, and ARM TrustZone dominate operational TEE deployments; EU alternatives (Keystone [71], OpenTitan [70]) are at research or prototype stage. Target: end-to-end privacy-preserving AI pipelines with progressive migration to EU-origin hardware. P9: Current infrastructures assume persistent connectivity. Target: validated ≥ 72 h Island Mode across NIS2-essential and DORA-critical deployments.

b: D2 Primary Drivers.

Threat: state-sponsored exploitation of debug interfaces and side-channel attacks on field-deployed nodes; supply chain insertion; “Harvest Now, Decrypt Later” campaigns; cascading failures from cyberattacks, extreme weather, and geopolitical tensions. Regulatory: CRA Articles 10–11 with enforcement from December 2027 [7]; NIS2 Article 21(2)(e) on cryptographic policies [6]; AI Act Articles 9 and 15 on hardware integrity for high-risk systems [8]; NIS CG PQC roadmap [30]; DORA Article 11 on business continuity [9]; SNS JU SRIA on 6G resilience [52].

c: D3 High-Impact Activities.

Phase 1: fund three to five RISC-V secure element pilots under Horizon Europe Cluster 3 targeting EUCC “High” pre-evaluation; submit OpenTitan and Keystone prototypes; optimise FIPS 203/204/205 for 8/16-bit microcontrollers; deploy hybrid ML-KEM plus ECDH-P384 on the cloud side per ETSI TS 103 744 [73]; validate Island Mode soft-state protocols in 5G testbeds. Phase 2: first EUCC “High” certified EU-manufactured RISC-V secure elements in production via Chips Act pilot lines (NanoIC, FAMES); RISC-V Keystone TEEs in multi-provider Data Spaces; certified 72 h Island Mode under 3GPP/ETSI fallback profiles in cross-border 5G corridors; mandatory hybrid PQC in DORA-scoped deployments. Phase 3: EU-designed RISC-V secure elements as the hardware default; FHE end-to-end from IoT sensor to HPC compute node; zero-energy PHY authentication for battery-less mMTC; mandatory validated 72 h autonomous Island Mode for NIS2-essential and DORA-critical deployments.

d: D4 Catalysts and Enablers.

EU Chips Act pilot lines (NanoIC, FAMES, APECS, PIXEurope, WBG; combined funding $\approx \text{€}3.7\text{B}$ [75]); DARE SGA1 (EuroHPC, $\text{€}240\text{M}$) [72]; ECCS Strategic Agenda [77]; CEN/CENELEC JTC 13 hardware security standards; NIST FIPS 203/204/205 [45]–[47] and ETSI TS 103 744 [73]; SNS JU 6G resilience projects [52].

TABLE 1. Focus areas per Security Layer and Cross-Layer.

Code	Focus area	Primary references
L1.1	Hardware Roots of Trust; EU open-hardware (RISC-V) to EUCC “High”	[22], [23], [70]–[72]
L1.2	Physical-layer threat detection and RF fingerprinting	[23]
L2.1	OS and middleware hardening; sovereign edge Linux	[39]
L2.2	Secure OTA patch and update management	[35], [36]
L3.1	Workload security (service mesh, mTLS, sidecar)	[37], [38]
L3.2	In-band network telemetry and deep observability	[38]
L3.3	API security gateway and policy enforcement	[39], [40]
L4.1	Zero-Trust orchestration (PEP/PDP)	[4], [5]
L4.2	Federated IAM and SSI	[41]–[43]
L4.3	Privileged Access Management and JIT	[4]
L5.1	Confidential computing and TEEs	[44]
L5.2	Post-Quantum Cryptography for edge/IoT	[30], [45]–[47], [73]
L5.3	Signalling/data-plane cryptographic authentication	
L5.4	Distributed reputation and trust networking (DLT)	
L5.5	QKD for critical links	[48], [49]
L5.6	Secure Multi-Party Computation	
L6.1	Telco-cloud and O-RAN security	[50]–[52], [74]
L6.2	Physical resilience and IT/OT safety	
L6.3	Edge-native autonomous guardians	
L6.4	Non-terrestrial network and satellite security	[53]
CL1.1	Continuous DevSecOps	[39]
CL1.2	Model integrity and DataBOMs	[54], [55]
CL1.3	Dynamic supply chain verification; Runtime SBOMs	[56]
CL1.4	Automated adversarial testing and purple teaming	[57]
CL2.1	Automated SOC and collaborative CTI	[58]
CL2.2	High-precision telemetry and security signalling	
CL2.3	Internet of Behaviour and evidence sharing	
CL2.4	AI-native analytics (NWDAF, LLMSecOps)	[59]
CL2.5	CTI lifecycle management	[60]–[63]
CL2.6	Deception technology and active defence	
CL3.1	Simpl integration and heterogeneous interoperability	[64]
CL3.2	Neutral host and multi-tenant isolation	
CL3.3	Pan-European NaaS federation	[65]
CL3.4	Semantic security interoperability	[66], [67]
CL3.5	Cyber range and digital twin federation	
CL4.1	PQC readiness and Zero-Trust architectures	[4], [5]
CL4.2	Continuous automated certification (EUCC; EUCS)	[10], [11], [68], [69]
CL4.3	Multicloud Key Management as a Service	
CL4.4	Privacy-preserving federated computation	[25]
CL4.6	Cyber risk quantification and board-level reporting	
CL4.7	AI trustworthiness and algorithmic accountability	[8], [54]

e: D5 Disruptors and Systemic Risks.

EU Chips Act fabrication capacity shortfall: ECA Special Report 12/2025 projects $\approx 11.7\%$ global semiconductor share by 2030 against the 20% target [76]; the cancellation of Intel Magdeburg in July 2025 removed the largest EU-based advanced node commitment. Early arrival of a Cryptographically Relevant Quantum Computer before migration is complete. PQC computational overhead on battery-powered IoT (ML-KEM at $\approx 1.5 \times$ ECDH on Cortex-M4 at equivalent security [78]). Side-channel breakthroughs against current TEE architectures. Insufficient cryptographic inventory of legacy OT/IoT systems.

f: D6 Failsafe Mechanisms.

TEE abstraction layers to avoid vendor lock-in during the Phase 1 dependency period; hybrid cryptographic schemes (classical plus PQC) for backward compatibility and harvest resistance; crypto-agility frameworks enabling sub-second algorithm switching; diversification across PQC algorithm

families (lattice, code, hash) to avoid single-point cryptanalytic failure; autonomous node quarantine via eBPF network policy if attestation fails; layered lightweight cryptography with PHY signatures as last-resort authentication under energy starvation; redundant trust anchors, segmented autonomy zones, and tamper-resistant local decision logic for Island Mode.

g: D7 Transversal Meta-Layer.

Sovereignty: eliminates proprietary hardware dependencies and quantum decryption exposure; current TEE dependency approaches 100%. Trust and Ethics: EUCC “High” aligns with Common Criteria EAL4+, AVA_VAN.5; PQC alignment with GDPR Article 32 [25]; the 18-Member-State joint statement “Securing Tomorrow, Today” (November 2024) formalises PQC as a collective sovereignty concern. Sustainable Security: energy-efficient secure silicon for far-edge; lightweight PQC extends device lifecycles and avoids premature replacement.

TABLE 2. Cluster 1 transition matrix, Sovereign Trusted Infrastructure.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D to Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	RISC-V secure element pilots in three to five Horizon Europe consortia; Intel/AMD/ARM TEEs dominate all operational deployments; classical cryptography on far-edge IoT; Island Mode validated in 5G testbeds by SNS JU consortia.	First EUCC “High” EU-manufactured RISC-V secure elements in production from Chips JU pilot lines; deployed in energy substations and telecom critical infrastructure in ≥ 5 Member States; RISC-V Keystone TEEs operational in multi-provider Data Spaces alongside Intel TDX; 72 h Island Mode certified in cross-border 5G corridors.	EU-designed secure elements as default in new EU ICT products; $\geq 60\%$ of critical infrastructure edge nodes on EUCC “High” EU-designed elements; FHE end-to-end across continuum; 72 h autonomous Island Mode mandatory in NIS2-essential and DORA-critical deployments.
Critical Enabler	EUCC “High” certification of an EU-origin open-source RISC-V secure element before end of phase.	Ratification of ETSI TS profiles for FIPS 203/204/205 tailored to EU deployment contexts; operational Chips JU fabrication capacity at the advanced nodes required for secure elements.	Mandatory EUCC “High” without waiver for new EU ICT products; PQC-native silicon as EU procurement default.

B. CLUSTER 2: COGNITIVE SECURITY AND AUTONOMOUS CYBER DEFENCE

Cluster 2 encompasses P6 (LLMSecOps) and P10 (Proactive APT Management), with cross-layer dependencies on Model Integrity and DataBOMs (CL1.2), Dynamic Supply Chain Verification (CL1.3), and AI Trustworthiness (CL4.7). The cluster is structured by three inseparable sub-dimensions: AI as a security instrument (sovereign models, edge-native agents, autonomous SOC, Moving Target Defence); AI as adversary amplifier (polymorphic malware, automated exploit discovery); AI as target (training pipelines and inference endpoints as attack surfaces under AI Act Article 9).

a: D1 Baseline and Target.

P6: US-platform SIEMs dominate the EU market, no EU-headquartered vendor in the Gartner Leaders quadrant [28]. Sovereign EU security-specific foundation models are absent; OpenEuroLLM [79] has no dedicated security stream. Target: cognitive SecOps model with EU-sovereign models, auditable under AI Act Article 9, covering $\geq 50\%$ of new critical infrastructure SOC deployments by 2040. P10: Average APT dwell times in European organisations are 150–200 days. Target: ≤ 72 h dwell time in Phase 2 and ≤ 24 h in Phase 3 across NIS2-essential deployments.

b: D2 Primary Drivers.

Threat: weaponisation of generative AI for polymorphic malware, automated exploit discovery, and scaled social engineering; nation-state actors blending espionage with organised crime; memory-only execution and abuse of legitimate administrative tools. Regulatory: AI Act Articles 9, 13, and 15 creating mandatory requirements for autonomous security AI classified as high-risk [8]; NIS2 Article 21(2)(b) on incident handling [6]. Conformity assessment capacity lags obligations by approximately 24–36 months [14].

c: D3 High-Impact Activities.

Phase 1: integrate commercial LLMs into SOC workflows for alert triage and MITRE ATT&CK mapping; route NWDAF telemetry into centralised AI analytics in three to five 5G SA pilots [59]; train a first EU-sovereign SLM (≤ 10 B parameters) on GDPR-compliant security corpora using EuroHPC [80]; apply ML-based UEBA for APT IoC detection; target sub-72 h dwell time in one certified NIS2-essential pilot. Phase 2: deploy EU-sovereign SLMs at the edge for real-time SOC analytics; deception infrastructure (honeynets, credential canaries) for calibration; autonomous triage under ETSI GS ZSM 009 [81]; Zero-Trust reconfigurations per NIST SP 800-207 [4]. Phase 3: swarm-based agentic AI for decentralised threat intelligence; continuous Moving Target Defence (IP, topology, software diversity); continuous automated adversarial robustness validation.

d: D4 Catalysts and Enablers.

EuroHPC AI Factories (19 confirmed across 16 Member States; cumulative funding $> \text{€}2.7\text{B}$; IT4LIA, PIAST, MeluXina-AI list cybersecurity as a vertical) [80]; ETSI GS ZSM 009 [81]; ECCC Strategic Agenda [77]; MITRE ATT&CK [60] and STIX/TAXII 2.1 [61], [62]; ENISA threat landscape reporting [63]; NIST AI RMF and Generative AI Profile [54], [55].

e: D5 Disruptors and Systemic Risks.

Hallucinations and automation bias in safety-relevant contexts; adversarial prompt manipulation and data leakage into models; absence of a standardised adversarial robustness test suite for security AI (the single most consequential standards gap in the cluster); intelligence fragmentation and weak information-sharing incentives; adversarial adaptation outpacing defensive innovation; CA body capacity shortfall [14].

f: D6 Failsafe Mechanisms.

Human-in-the-loop validation for Tier-1 critical actions; bounded autonomy governed by ETSI GS ZSM 009 [81]; shadow-mode rollback if autonomous agents exhibit unstable control loops; policy-constrained orchestration with audit logging; autonomous real-time micro-segmentation to quarantine compromised components under Zero-Trust; layered detection, deception, and rapid recovery playbooks.

g: D7 Transversal Meta-Layer.

Sovereignty: reduces approximately 80% US-platform dependency in enterprise SIEM. Trust and Ethics: AI Act Article 9 compliance-by-design, explainability, and accountability for autonomous security decisions; NIS2 sector-specific implementing acts should reference MITRE ATT&CK [60]. Sustainable Security: edge-native SLMs reduce inference-time energy cost versus cloud round-trips for high-volume security telemetry.

C. CLUSTER 3: FEDERATED OPERATIONAL SOVEREIGNTY

Cluster 3 encompasses P4 (Security Interoperability via Simpl), P5 (Supply Chain Verification), and P7 (Pan-European NaaS Federation). The cluster addresses three mechanisms that transform sovereign components into a collectively sovereign continuum: federated interoperability through Minimal Interoperability Mechanisms (MIMs) in the Simpl stack [64] under Data Act portability mandates [24]; supply chain integrity through Runtime SBOMs, Digital Product Passports, and cryptographic proof-of-execution under CRA Annex I [7]; and compute federation through harmonised CAMARA Network APIs [65] and cross-border edge roaming. Supply chain verification bridges Clusters 1 and 3 because hardware attestation depends on the RISC-V secure elements matured in Cluster 1.

a: D1 Baseline and Target.

P4: European cloud-edge ecosystems remain fragmented across proprietary vendor silos with incompatible security policies. The Simpl-Open iterative MVP has been available since January 2025, with enhanced releases in June and December 2025 [64]. OASC MIM6 (Security management) remains a “Work item” without a formally published specification. Target: unified European trust fabric based on Simpl MIMs ratified as CEN/CENELEC standards and referenced in the EU Cloud Rulebook. P5: Current supply chains are opaque and globally distributed. CRA Annex I Part II mandates static SBOM disclosure from December 2027 [7]; NIS2 Article 21(2)(h) mandates supply chain risk assessment [6]; DORA Article 28 requires continuous ICT third-party monitoring [9], operationalised through the first 19 Critical ICT Third-Party Provider designations on 18 November 2025 [82]. Target: progression from static SBOM disclosure (Phase 1) to Runtime SBOMs with cryptographic proof-of-execution (Phase 2) to ZKP-verified supply chain attestation at transaction scale (Phase 3). P7: The CAMARA

Fall 2025 meta-release comprises 60 APIs (10 stable) [65]; GSMA Open Gateway reaches 86 operator groups [83]. Cross-border edge compute roaming does not exist at operational scale. Target: commercial EU NaaS federation as a unified jurisdiction for data workloads, with non-EU hyper-scaler market share in EU regulated cloud procurement below 40% for new deployments by 2040.

b: D2 Primary Drivers.

Threat: SolarWinds-class software supply chain attacks; disaggregation of telco clouds introducing multi-vendor complexity; the sovereignty cliff where absence of a Digital Single Market prevents EU operators from scaling against US and China hyperscalers. Regulatory: Data Act (applicable from 12 September 2025) [24]; NIS2 coordinated incident response [6]; CRA SBOM obligations (December 2027) [7]; DORA CTPP monitoring [9]; the Digital Networks Act proposal (COM(2026) 16, January 2026) signals EC intent for network federation but does not mandate API standards.

c: D3 High-Impact Activities.

Phase 1: Simpl security MIM pilots in three Common European Data Spaces (Manufacturing, Health [84], Energy); deploy harmonised CAMARA APIs in two to three bilateral 5G corridor agreements; submit CRA-mandated static SBOM declarations; initiate Digital Product Passport pilots; advance ZKP verification towards demonstrator readiness. Phase 2: automated policy translation via standardised Simpl MIMs across active Data Spaces; Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in ≥ 5 Member States; automatic quarantine of unverified components in isolated network slices; CAMARA cross-border roaming scaled to ≥ 5 operator pairs; “Data Visa” mechanisms for dynamic AI model migration. Phase 3: Simpl-based security as universal Data Space standard; ZKP-verified supply chain attestation at real-time scale; operational EU NaaS federation; “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict.

d: D4 Catalysts and Enablers.

Digital Europe Programme funding for Simpl [64]; CEN-CLC/JTC 25 for data-space standards; Eclipse Dataspace Protocol; IPCEI-CIS ICRA Reference Architecture v2.0 (January 2026, total €2.6B, 19 participants, 12 Member States) [85]; EU Ecodesign Regulation Digital Product Passport; DORA Delegated Regulation (EU) 2024/1502 for CTPP assessment [9]; SNS JU [52]; Aduna as primary aggregator.

e: D5 Disruptors and Systemic Risks.

Vendor resistance to open standards; weakest-link failures in federated systems; uneven Member State compliance maturity; US CLOUD Act jurisdictional conflict [26]; concentration of manufacturing dependencies; limited multi-tier auditability; SME exclusion from compliance pathways;

TABLE 3. Cluster 2 transition matrix, Cognitive Security and Autonomous Cyber Defence.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Applied R&D, Siloed AI Copilots	Standardised Integration	Ubiquitous Operation
SSL	High Reliance on US Foundation Models	Sovereign EU Edge LLMs Emerging	Global Leadership in Trustworthy AI Security
Operational State	Commercial LLMs integrated into SOC workflows for triage; NWDAF pilots in three to five 5G SA deployments; first EU-sovereign SLM under training on EuroHPC AI Factory compute; sub-72 h dwell time achieved in at least one certified pilot; no operational autonomous containment agent.	EU-sovereign SLMs deployed at the edge for real-time SOC analytics without cloud inference; autonomous triage governed by ETSI GS ZSM 009 in production; sub-72 h dwell time in certified critical infrastructure deployments; deception infrastructure operational for calibration.	Swarm-based agentic AI defending the continuum via self-evolving decentralised CTI; Moving Target Defence continuous in certified environments; ≤ 24 h dwell time across NIS2-essential deployments; EU-sovereign AI security tools in $\geq 50\%$ of new critical infrastructure SOC deployments.
Critical Enabler	CEN/CENELEC or ETSI TC CYBER publication of a standardised adversarial robustness test suite for security AI before end of phase.	Operationalisation of an AI Act Article 9 conformity assessment framework specific to autonomous security agents; activation of EuroHPC AI Factory compute contracts for EU-sovereign security model training.	International adoption of EU AI Act frameworks for trustworthy autonomous security agents by at least one non-EU partner jurisdiction.

hyperscaler platform dominance through proprietary API ecosystems (approximately 70% EU cloud share) [29].

f: D6 Failsafe Mechanisms.

Autonomous unfederation or isolation of nodes failing the shared baseline; Zero-Trust architectures where federation does not imply implicit trust [4]; cryptographic attestation of policy enforcement; graceful isolation of unverified components into quarantined network slices; multi-vendor diversification; trusted fallback components; graceful degradation redirecting workloads to national infrastructure; regional autonomy zones; multi-operator fallback routing.

g: D7 Transversal Meta-Layer.

Sovereignty: breaks hyperscaler lock-in and resolves structural CLOUD Act conflict. Trust and Ethics: transparent and verifiable federation mechanisms enabling cross-border trust; the 2025 DINUM Bilan reports 70% European-provider share in French public cloud procurement market-wide and 99% within the State perimeter [86]. Sustainable Security: shared infrastructure reduces duplication of compute and certification overhead across Member States.

D. CROSS-CUTTING ENABLERS

Four dependencies span at least two clusters and require dedicated governance to avoid becoming systemic bottlenecks.

EU Chips Act sovereign fabrication (Clusters 1 and 3). Five Chips JU pilot lines are operational, with combined EU and national funding of €3.7B [75]. No pilot line is dedicated to secure element fabrication; restriction of access to any major non-EU foundry would immediately impair EU critical infrastructure attestation capability. Mitigation: strategic stockpiling, second-source agreements, and dedicated pilot lines for security-critical components.

AI Act conformity assessment infrastructure (Cluster 2). A 24–36 month gap between AI Act obligations and CA body operational readiness is documented [14]. As of March 2026, 8 of 27 Member States have designated single points of

contact. Mitigation: accelerated CEN/CENELEC adversarial robustness test suite development and AI Act sandbox provisions for security use cases.

Runtime SBOM attestation (Clusters 1 and 3). Static SBOMs (CRA-mandated) are insufficient for dynamic continuum operations. Progression to cryptographic proof-of-execution and ZKP batch verification is a shared Phase 2–3 transition condition.

SME access and inclusion (all clusters). Cost barriers exclude SMEs from sovereign hardware alternatives, AI-powered SOC tooling, and Simpl MIM compliance. Digital Europe Programme co-funding should target shared-infrastructure pathways and subsidised certification programmes.

VII. STRATEGIC RECOMMENDATIONS

The 2040 destination is a secure, sovereign, interoperable, and resilient European computing continuum. Six actions follow from the analysis.

Sovereign secure silicon. Combine Horizon Europe support with the EU Chips Act to advance open-source RISC-V secure elements and pilot fabrication lines reaching EUCC “High” assurance. The 2025 certifications achieved by Infineon, NXP, and STMicroelectronics [27] demonstrate framework readiness; the remaining gap is open-source, EU-controlled fabrication.

Operationalised PQC migration. Deliver crypto-agility middleware for edge and IoT aligned with the NIS CG PQC roadmap milestones [30] and ETSI TS 103 744 hybrid key exchange profiles [73]. Make hybrid cryptographic deployment the default for new critical infrastructure.

Security interoperability through Simpl. Move OASC MIM6 (Security management) from Work item status to a published specification, then to a ratified CEN/CENELEC standard referenced in the EU Cloud Rulebook, so that interoperability carries procurement-level enforcement.

Continuous certification. Use EUCC [11] as the foundation and unblock EUCS [10], plausibly through the 2026

TABLE 4. Cluster 3 transition matrix, Federated Operational Sovereignty.

	Phase 1 (2026–2030)	Phase 2 (2030–2035)	Phase 3 (2035–2040)
ODP	Fragmented Piloting	Standardised Integration	Ubiquitous Operation
SSL	Mixed: Emerging EU Ecosystem / Hyper-scaler Dominance	Emerging EU Ecosystem	Full Federated Sovereign Autonomy
Operational State	Simpl security MIM pilots in three Common European Data Spaces with manual policy coordination; CAMARA APIs in two to three bilateral 5G corridors; no operational cross-border edge compute roaming; CRA-mandated static SBOMs submitted as point-in-time artefacts; DPP pilots initiated; hyperscalers retain approximately 70% of EU cloud market.	Automated policy translation via standardised Simpl MIMs across active Data Spaces; Runtime SBOMs with cryptographic proof-of-execution in critical infrastructure in ≥ 5 Member States; ZKP batch verification operational; CAMARA roaming across ≥ 5 operator pairs; “Data Visa” mechanisms operational; non-hyperscaler share $\geq 30\%$ in new regulated-sector procurement.	Simpl-based security as universal cross-provider Data Space standard; ZKP-verified supply chain attestation at real-time scale; operational EU NaaS federation; “Schengen for Cloud” framework enacted; non-EU hyperscaler share below 40% in EU regulated cloud procurement for new deployments; EU supply chain standards adopted in ISO/IEC and by at least two partner jurisdictions.
Critical Enabler	Simpl-Open iterative MVP consolidation with OASC MIM6 (Security management) progressing from Work item to formally published specification.	Ratification of Simpl security MIMs as CEN/CENELEC standards referenced as mandatory in the EU Cloud Rulebook; completion of DORA continuous ICT third-party monitoring operationalisation.	Enactment of the “Schengen for Cloud” legal framework resolving US CLOUD Act jurisdictional conflict.

Cybersecurity Act revision (COM(2026) 13) [13], to extend continuous assurance to cloud services.

Federated SOC and CTI. Build on DORA CTPP oversight [82] and cross-border CSIRT cooperation [58] to support pan-European situational awareness.

SME inclusion mandate. Implement tiered certification, shared cyber range access, and reusable compliance tooling so that NIS2, CRA, and AI Act compliance does not become a structural barrier to innovation.

VIII. DISCUSSION

A. VALIDATION

The roadmap is treated as an evolving strategic artefact benefiting from expert review, stakeholder consultation, and iterative revision. Validation is conducted through expert workshops within the NexusForum and EuCloudEdgeIoT Working Group on Cybersecurity; consultation with national cybersecurity agencies and industry; and policy dialogue with European Commission services. Validation addresses completeness and relevance of the architectural decomposition, clarity and non-overlap of capability formulations, coherence of priority clustering, credibility of transition conditions, and reasonableness of Sovereignty Level assessments.

B. LIMITATIONS

The architectural decomposition reflects a particular analytical perspective and may not capture intersections with adjacent domains (energy, industrial policy, trade). The seven-dimension profiles rely on expert judgement rather than empirical measurement. Transition conditions sacrifice precision for robustness; they are suited to strategic orientation rather than operational planning. The Sovereignty Level construct resists precise quantification and is used interpretively. The paper also reflects the ambition–implementation gap of the 2026 regulatory landscape: NIS2 transposition delays in several large Member States [12], a contested 20% semiconductor target [76], EUCS remaining unadopted [13], and no

harmonised CRA standards yet cited in the Official Journal. These realities constrain Phase 1 governance specificity and locate the roadmap’s primary value in structuring coordination.

C. METHODOLOGICAL POSITIONING

The methodology is suited to strategic agenda setting, expert discussion, cross-priority comparison, and iterative roadmap evolution under conditions of ecosystem uncertainty. Its value lies in providing a shared analytical structure within which stakeholders can coordinate research, policy, and investment decisions toward a common set of objectives.

IX. CONCLUSION

The Cloud–Edge–IoT continuum requires Europe to move beyond reactive and fragmented cybersecurity models towards autonomous, federated, and certified Security-by-Design. The structured methodology presented here identifies security dimensions, derives strategic capabilities, and consolidates them into priority clusters aligned with resilience and digital sovereignty objectives. The ten capabilities form an interdependent strategic portfolio that must be developed in parallel. The 2025–2026 period has produced concrete milestones (EUCC “High” certifications across all three major EU secure element manufacturers, a binding PQC migration framework, DORA operationalisation through CTPP designation, and emerging Simpl infrastructure) and persistent gaps (NIS2 transposition delays, EUCS deadlock, contested semiconductor targets, continued hyperscaler dominance at approximately 70% of the EU cloud market). The ambition–implementation gap defines where the coordination challenge is most acute and where strategic investment will yield the greatest returns. Future work should extend this analysis through empirical validation of Sovereignty Level assessments, sector-specific instantiation (energy, healthcare, financial services), and integration with the next EU Framework Programme funding landscape.

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